

SECOND PUBLIC EXAMINATION

Honour School of Physics Part B: 3 and 4 Year Courses

Honour School of Physics and Philosophy Part B

B3: ATOMIC AND LASER PHYSICS

TRINITY TERM 2025

Tuesday, 10 June, 2:30 pm – 4:30 pm

*Answer **two** questions.*

*Start the answer to each question in a **fresh book**.*

The use of approved calculators is permitted.

A list of physical constants and conversion factors accompanies this paper.

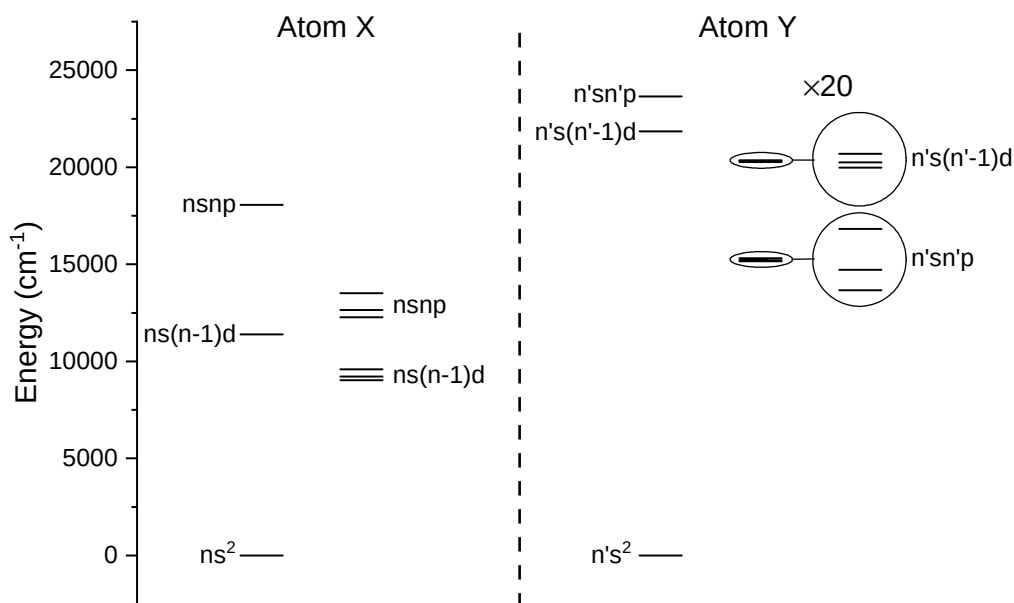
The numbers in the margin indicate the weight that the Examiners expect to assign to each part of the question.

Do NOT turn over until told that you may do so.

1. (a) Explain what is meant by the central field approximation and discuss how it leads to the concept of electron configurations. [4]

(b) Qualitatively explain (no detailed derivations are required) the physical processes that lead to splitting of a configuration into terms and levels. [3]

The level diagrams below show the lowest 3 configurations of Calcium ($Z=20$) and Barium ($Z=56$) with the energies measured relative to the ground configuration.



(c) Identify which atom (X or Y) is Ca and which is Ba; give your reasoning. [3]

(d) On copies of the energy level diagrams, label all the levels with the appropriate symbols, including the values of n and n' . [5]

(e) Explain the relative positions of the triplet and singlet levels. [2]

(f) State what is meant by the 'interval rule'. For which of Ba or Ca would you expect it to be most closely obeyed? Give your reasoning. [4]

(g) For atom X, list all the allowed electric dipole transitions. Suggest, with reasons, which is likely to be the strongest line. [4]

2. (a) Write down the characteristic energy scales associated with gross, fine, and hyperfine structure in a hydrogen atom in terms of the fine structure constant α , together with the speed of light c , and any relevant masses. Briefly explain the physical origins and the relative sizes of the spin-orbit and hyperfine interactions (you only need consider $Z = 1$). [6]

(b) Positronium is a hydrogen-like system where the proton is replaced by a positron (the anti-particle of the electron, which has spin $1/2$ and the same mass as the electron but opposite charge). Apply the Bohr model to positronium to find an expression for its gross energy levels. For the $1s$ configuration, what is the binding energy, expressed in eV, and what is the distance between positron and electron, expressed in terms of the Bohr radius? [3]

For $1s$ positronium in an external magnetic field $\mathbf{B}$, the relevant Hamiltonian is

$$H = A\mathbf{s}_e \cdot \mathbf{s}_p + 2\mu_B(\mathbf{s}_e - \mathbf{s}_p) \cdot \mathbf{B} .$$

where $\mathbf{s}_e$ and $\mathbf{s}_p$ refer to the electron and positron spin respectively.

(c) How many eigenstates does this Hamiltonian have? What are the good quantum numbers in the low and high magnetic field regimes? [4]

(d) Find the energies of all the states in both the zero- and high-field limits. [5]

(e) Sketch the energy levels as a function of B , going up to the high-field regime; use appropriate annotations including the scale of B . [4]

(f) Briefly discuss how the concepts of fine and hyperfine structure apply to the p -configurations of positronium. [3]

3. A two-level atom with energy levels E_1 and E_2 is illuminated by a pulse of monochromatic radiation of length T ; the radiation has an electric field given by $\mathbf{E} = \mathcal{E}_0 \cos(\omega t) \hat{\mathbf{z}}$. Under certain assumptions, the population of the upper state at the end of the pulse is given by

$$|c_2(T)|^2 = \frac{\Omega^2}{\Omega^2 + \delta^2} \sin^2 \left(\frac{\sqrt{\Omega^2 + \delta^2} T}{2} \right). \quad (1)$$

(a) Give expressions for both Ω and δ in terms of the quantities given and any relevant matrix elements. [3]

(b) Outline the assumptions that are required to arrive at Eq. (1), for each give any relevant inequalities that need to be satisfied. [5]

(c) Sketch how the upper state population varies with T for both $\delta = 0$ and $\delta = \Omega$, labelling the relevant times and populations. [4]

(d) Compare the populations in these two cases in the limit $T \rightarrow 0$. What is the condition on T for this limiting case to apply? Comment on your result. [3]

The energy of the upper state (E_2) of the atoms is sensitive to any external magnetic field. An ensemble of these atoms is in a region of space with a distribution of magnetic fields which result in a range of possible E_2 values; in what follows, assume the distribution of E_2 values is uniform with full width $\hbar\Delta$.

(e) The radiation is tuned to be on resonance for atoms with the central (average) value of E_2 , what pulse length should be chosen to achieve a π -pulse for these atoms? [1]

(f) Under these conditions and assuming $\Delta \ll \Omega$, derive an expression for the ensemble averaged upper state population as a function of Δ and Ω . What Ω/Δ is required in order to achieve an upper state population of $> 99\%$? Comment on why this favours using high-power radiation to achieve high-fidelity π -pulses. [6]

(g) If instead T is chosen such that $\Omega T \gg 1$ and $\Delta T \gg 1$ (still with $\Omega \gg \Delta$), what upper state population will result? Give your reasoning. [3]

4. (a) Describe the difference between homogeneous and inhomogeneous line broadening. Give an example of each. [3]

Consider an ensemble of atoms which can be modeled as non-degenerate two-level systems with an energy splitting of $E_2 - E_1 = \hbar\omega_0$ and a spontaneous decay lifetime of the upper state τ . The populations of the two levels are N_1 and N_2 atoms per unit volume. The atoms are subject to a beam of monochromatic radiation of frequency ω and uniform intensity I which propagates along the z axis. The radiation leads to stimulated absorption and emission at rates (per unit volume per unit time) of $R_{12} = N_1\sigma(\omega - \omega_0)\frac{I}{\hbar\omega}$ and $R_{21} = N_2\sigma(\omega - \omega_0)\frac{I}{\hbar\omega}$. Here $\sigma(\omega - \omega_0)$ is the optical cross-section of the transition.

(b) Write down a rate equation for the population in the upper level. [2]

(c) In steady-state, how much optical power is absorbed per unit volume? (Hint: consider which of the processes lead to loss of power from a propagating beam.) Express your answer in terms of the steady-state population density difference ΔN . Use this result to write down an expression for $\frac{dI}{dz}$ which describes the change in intensity of the beam as it passes through the gas. [5]

(d) Find the steady-state population density difference ΔN in terms of the total population density N_T and use this to write an alternative expression for $\frac{dI}{dz}$ of the form $\frac{dI}{dz} = -\frac{\kappa I}{1+I/I_s}$; give expressions for κ and I_s . Describe the significance and physical origin of the saturation intensity I_s . [5]

Consider a dilute gas of rubidium atoms (atomic mass 85), with a number density of $N_T = 10^8 \text{ cm}^{-3}$, that is contained within a glass cell of length 10 cm. The rubidium has an atomic transition at $\omega_0 = 2\pi c/\lambda_0$ (with $\lambda_0 = 780 \text{ nm}$) with an excited state lifetime of $\tau = 27 \text{ ns}$. The peak natural absorption cross section at this transition is $3 \times 10^{-9} \text{ cm}^2$. A weak ($I \ll I_s$) laser beam of frequency ω , which is close to ω_0 , is shone through the cell and the transmitted intensity measured.

(e) For the two cases of a gas at temperature of 0.001 K and one at 10 K, draw a sketch of the transmitted intensities as a function of ω , in the region around ω_0 , with appropriate scales. For each case, state the line shape of the absorption feature and estimate its FWHM. [7]

(f) For the case of the 10 K gas, a stronger laser beam ($I \approx I_s$) of frequency ω_0 is now applied counter propagating to the weaker beam. Make a third sketch (on the same plot) showing the ω dependence of the transmitted intensity of the *weak* beam. Again explain your reasoning. [3]